



From algorithms to invention: AI's impact on corporate innovation output and efficiency

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ABSTRACT

This paper explores how artificial intelligence (AI) drives corporate innovation. Using a procurement-based AI adoption index, patent records, and supply-chain data from 4004 A-share firms over 2011–2023, we find that greater AI adoption significantly increases the output and efficiency of firms' innovation. We propose a dual-channel model in which AI enhances knowledge creation and reuse (knowledge orchestration) and transforms data into actionable environmental assets (data assetization). Heterogeneity analysis reveals that large incumbents and growth-stage firms leverage AI most effectively for innovation outputs and efficiency. Further analysis shows that AI-driven innovation is amplified in firms with executives who have information technology backgrounds, and that highly innovative firms diversify their supply chain to reduce resource risk. Our results demonstrate AI's potential to advance corporate innovation. We conclude with policy recommendations for municipal planners and corporate strategists to enhance firms' competitive advantage and promote the development of AI.

1. Introduction

Artificial intelligence (AI) has become a central driver of corporate innovation, reshaping how firms create new products, processes, and business models (Johnson et al., 2022; Liu et al., 2024a, 2024b). Facing intense competition and sustainability pressures, companies use AI not just for automation but to transform innovation from incremental improvement into a dynamic, systemic process (Ayman Wael Al-Khatib, 2023). This paper examines artificial intelligence's role in corporate innovation. We argue that AI's impact extends through enhanced efficiency, quality, and strategic repositioning across value chains.

A burgeoning body of literature has documented the transformative effects of AI on innovation paradigms. Bahoo et al. (2023) offer a comprehensive taxonomy that situates AI as a general-purpose technology capable of redefining corporate research and development (R&D) processes. Dong et al. (2025) and Zhong and Song (2025) show

that AI boosts both the quantity and quality of innovation, mediated by human capital, management, and financing constraints. Wang, Wang, et al. (2024) link digital technology to process innovation, while Guo and Xu (2024) analyze urban digital intelligence transformation, underscoring AI's wider societal and economic impact. Jiang et al. (2024) examine how AI reconfigures corporate innovation, and Kim et al. (2022) highlight its role in driving social innovation.

Despite these insights, important gaps remain. First, many studies adopt a unidimensional perspective by focusing predominantly on either innovation output or efficiency, thereby neglecting the dual-channel effects that AI may exert across multiple innovation dimensions. Second, conventional measurement proxies—often reliant on patent counts or self-reported AI adoption metrics—fail to capture the nuanced interplay between AI-enabled technological integration and complementary organizational assets such as data governance and executive technical expertise. Third, while prior work explores firm size and

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